SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Code of Conduct
I **__Lavanya.M(192321051)__** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: **__Lavanya.M__**

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Assessment Tasks

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

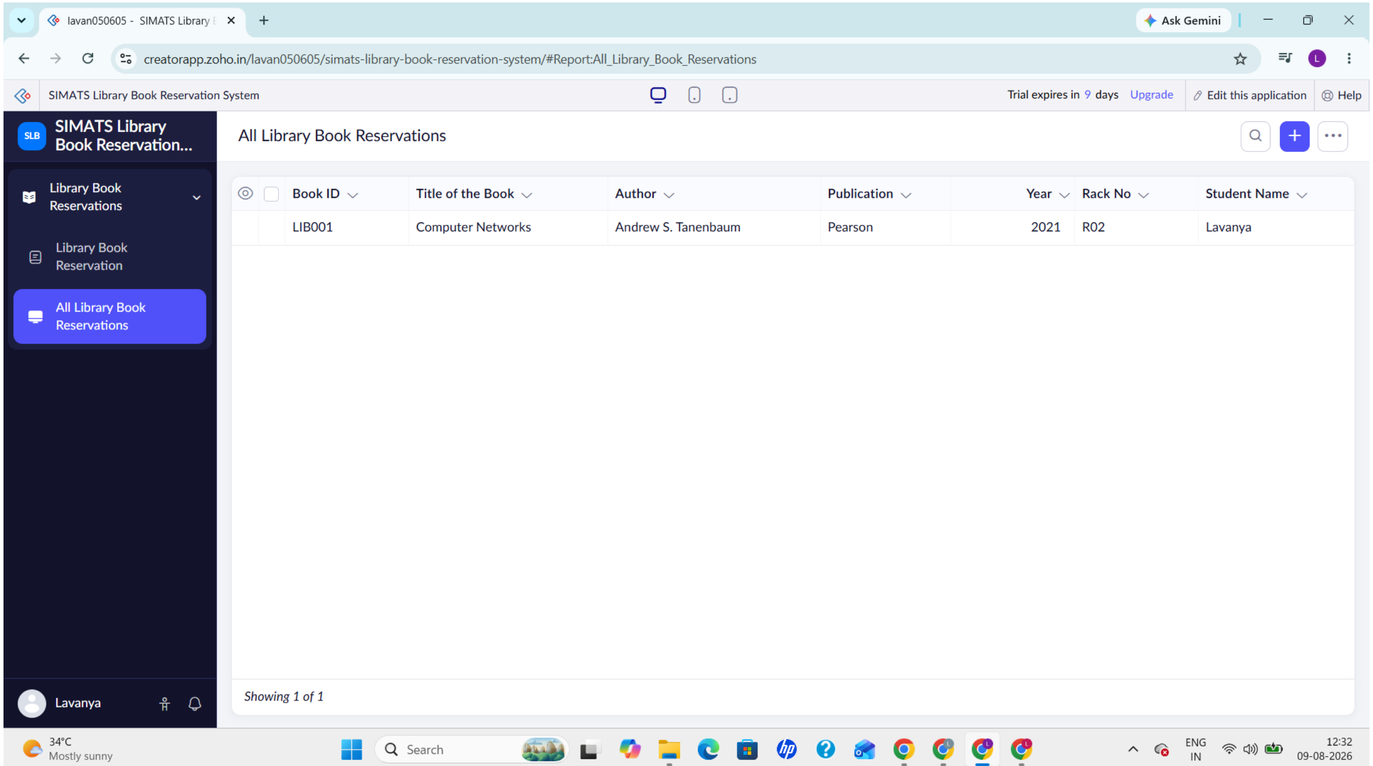
Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

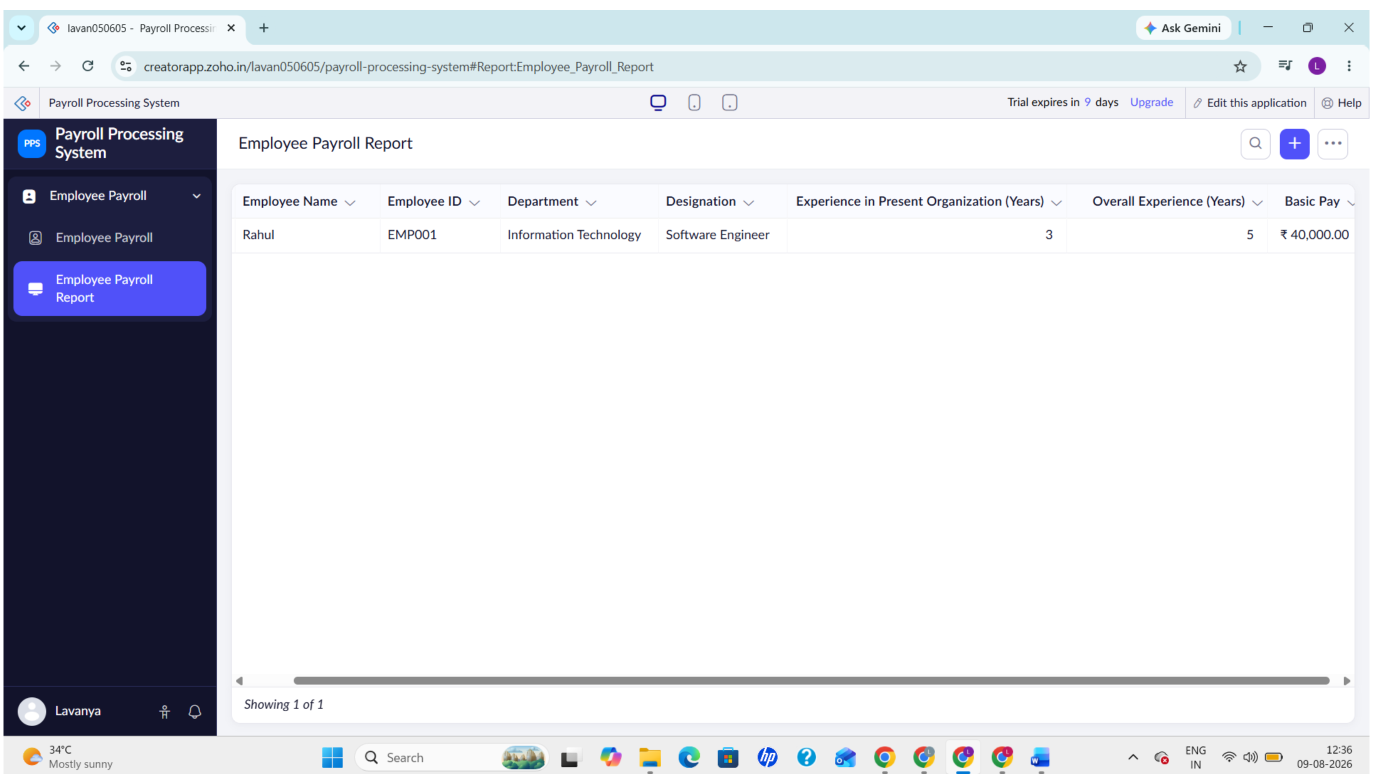
Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

ASSESSMENT TASK-1

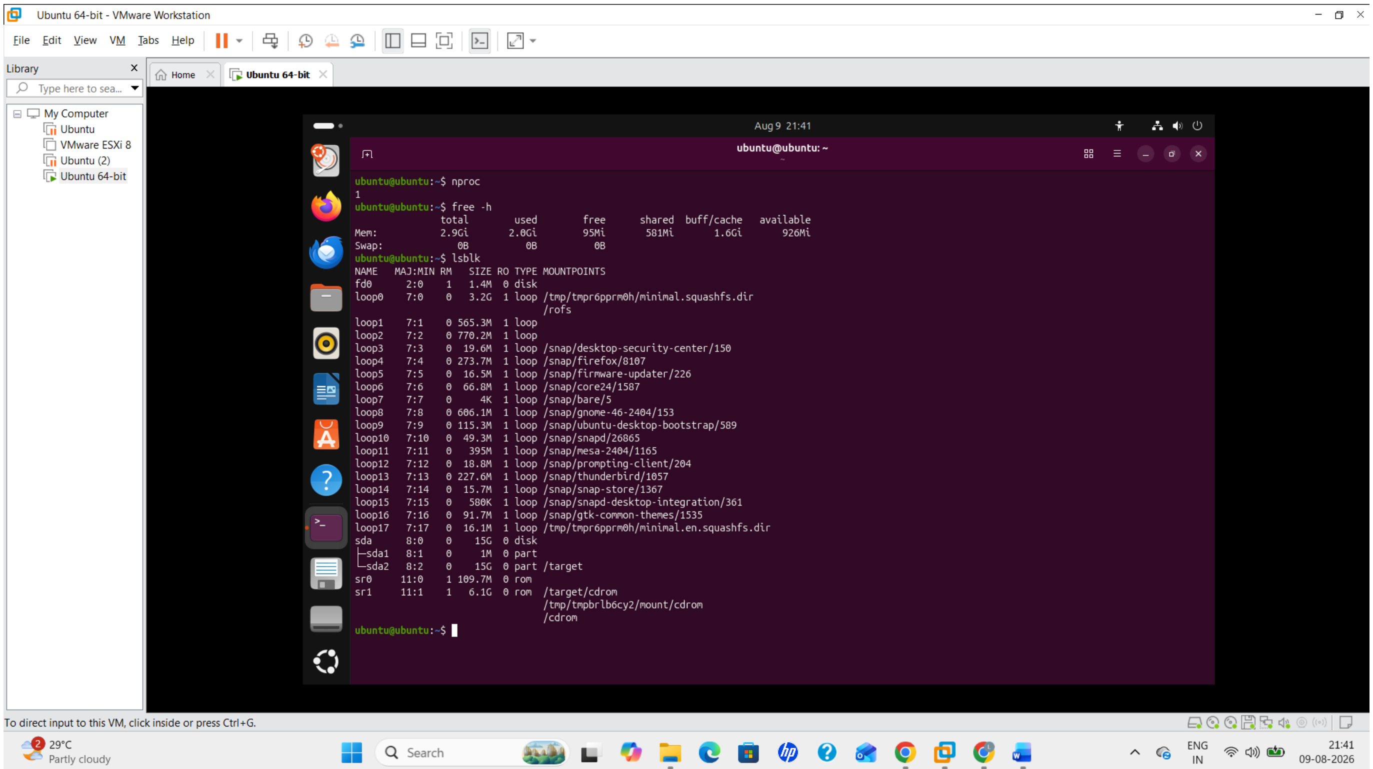
EXP 1:



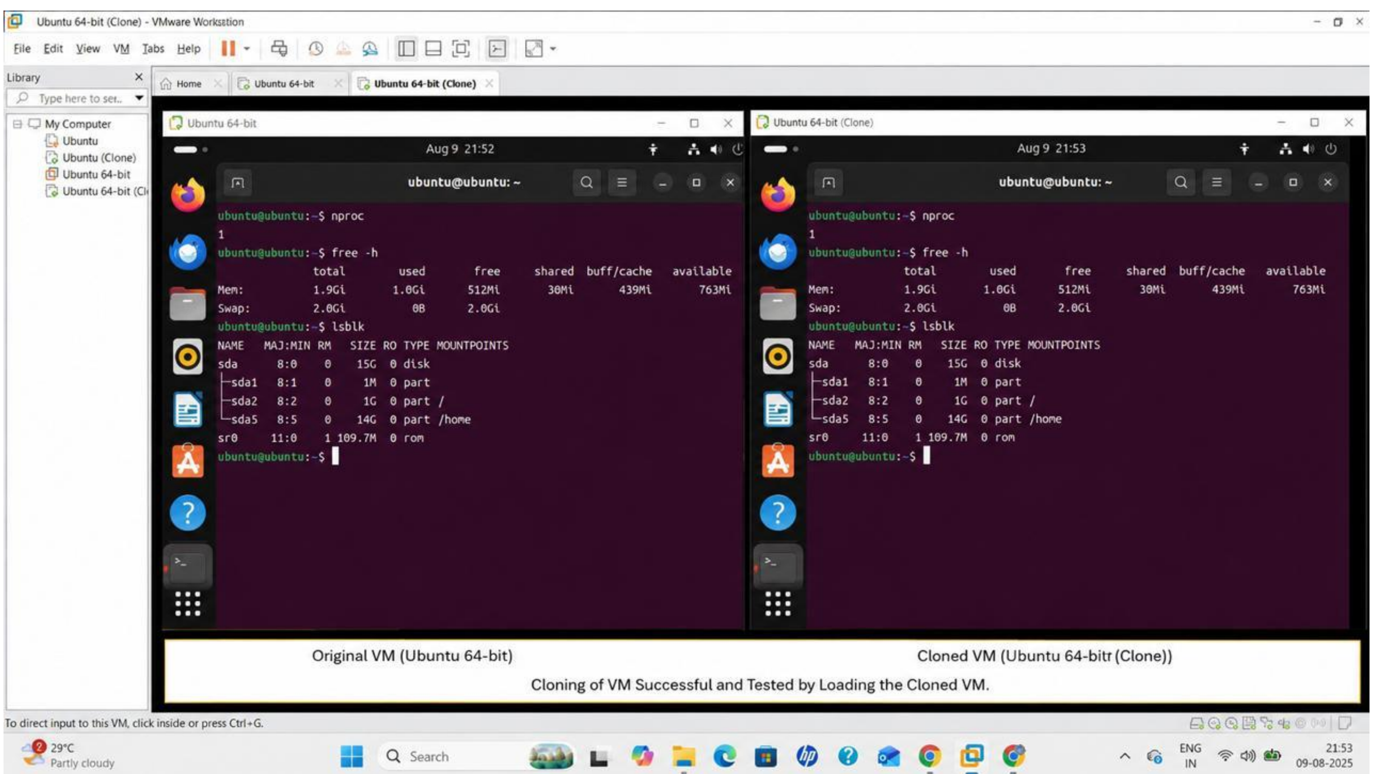
EXP 2:



EXP 3:

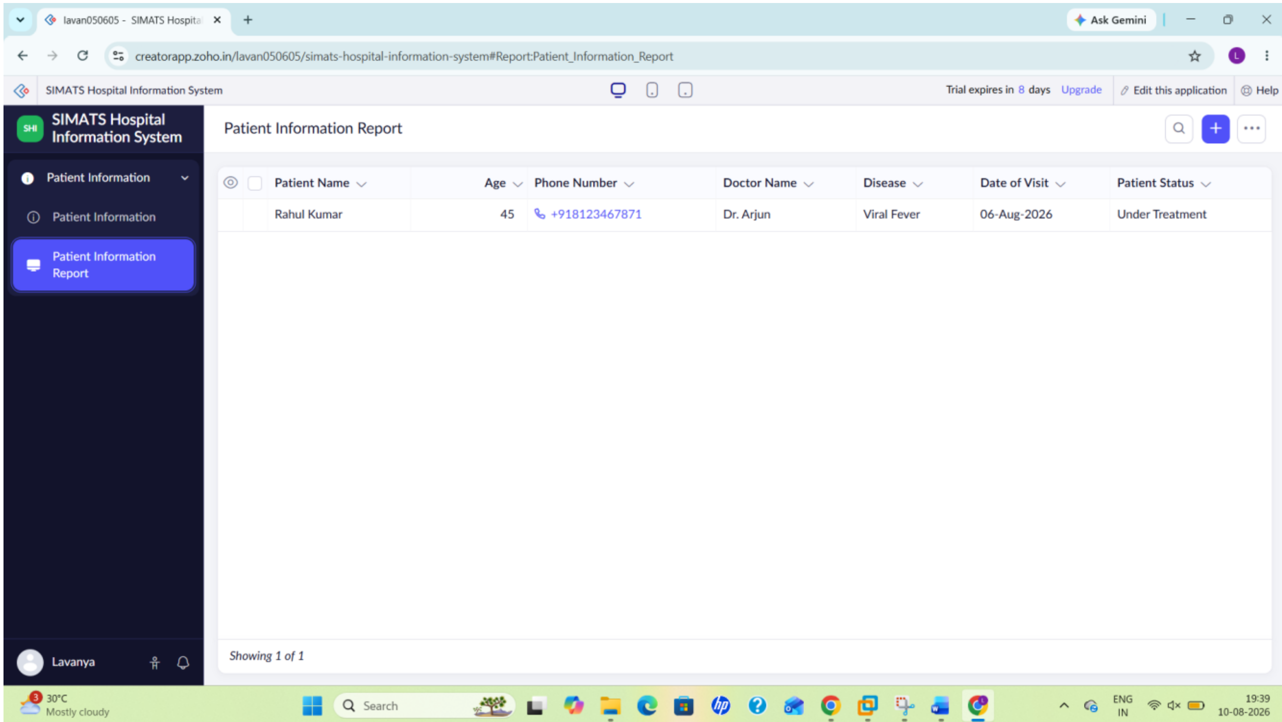

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu
VMware ESXi 8
Ubuntu (2)
Ubuntu 64-bit
Home
Ubuntu 64-bit
Aug 9 21:41
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ nproc
1
ubuntu@ubuntu:~$ free -h
total used free shared buff/cache available
Mem: 2.9Gi 2.0Gi 95Mi 581Mi 1.6Gi 926Mi
Swap: 0B 0B 0B
ubuntu@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
fd0 2:0 1 1.4M 0 disk
loop0 7:0 0 3.2G 1 loop /tmp/tmpr6pprm0h/minimal.squashfs.dir/rofs
loop1 7:1 0 565.3M 1 loop
loop2 7:2 0 770.2M 1 loop
loop3 7:3 0 19.6M 1 loop /snap/desktop-security-center/150
loop4 7:4 0 273.7M 1 loop /snap/firefox/8107
loop5 7:5 0 16.5M 1 loop /snap/firmware-updater/226
loop6 7:6 0 66.8M 1 loop /snap/core24/1587
loop7 7:7 0 4K 1 loop /snap/bare/5
loop8 7:8 0 606.1M 1 loop /snap/gnome-46-2404/153
loop9 7:9 0 115.3M 1 loop /snap/ubuntu-desktop-bootstrap/589
loop10 7:10 0 49.3M 1 loop /snap/snapd/26865
loop11 7:11 0 395M 1 loop /snap/mesa-2404/1165
loop12 7:12 0 18.8M 1 loop /snap/prompting-client/204
loop13 7:13 0 227.6M 1 loop /snap/thunderbird/1057
loop14 7:14 0 15.7M 1 loop /snap/snap-store/1367
loop15 7:15 0 580K 1 loop /snap/snapd-desktop-integration/361
loop16 7:16 0 91.7M 1 loop /snap/gtk-common-themes/1535
loop17 7:17 0 16.1M 1 loop /tmp/tmpr6pprm0h/minimal.en.squashfs.dir
sda 8:0 0 15G 0 disk
├─sda1 8:1 0 1M 0 part
└─sda2 8:2 0 15G 0 part /target
sr0 11:0 1 109.7M 0 rom
sr1 11:1 1 6.1G 0 rom /target/cdrom /tmp/tmpbrlb6cy2/mount/cdrom /cdrom
ubuntu@ubuntu:~$
To direct input to this VM, click inside or press Ctrl+G.
29°C
Partly cloudy
Search
ENG
IN
21:41
09-08-2026

EXP 4:

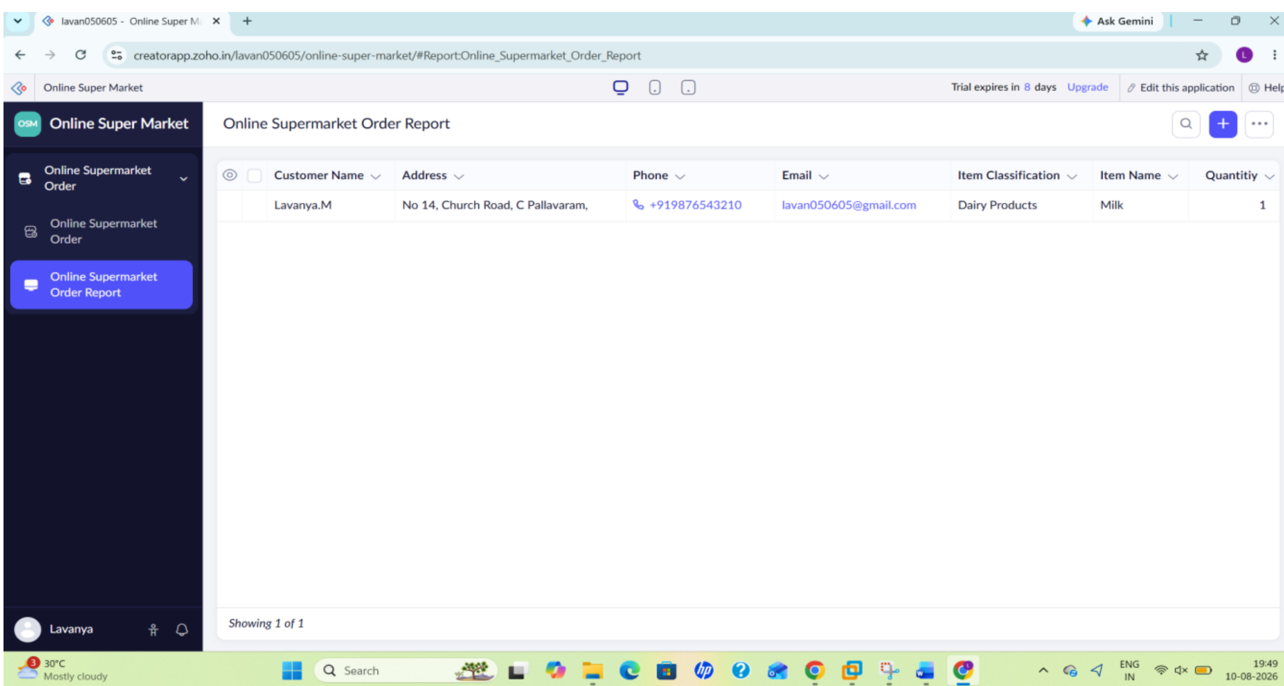

Ubuntu 64-bit (Clone) - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu
Ubuntu (Clone)
Ubuntu 64-bit
Home
Ubuntu 64-bit
Ubuntu 64-bit (Clone)
Aug 9 21:52
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ nproc
1
ubuntu@ubuntu:~$ free -h
total used free shared buff/cache available
Mem: 1.9Gi 1.0Gi 512Mi 30Mi 439Mi 763Mi
Swap: 2.0Gi 0B 2.0Gi
ubuntu@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 15G 0 disk
├─sda1 8:1 0 1M 0 part
├─sda2 8:2 0 1G 0 part /
└─sda5 8:5 0 14G 0 part /home
sr0 11:0 1 109.7M 0 rom
ubuntu@ubuntu:~$
Aug 9 21:53
Original VM (Ubuntu 64-bit)
Cloned VM (Ubuntu 64-bitr (Clone))
Cloning of VM Successful and Tested by Loading the Cloned VM.
To direct input to this VM, click inside or press Ctrl+G.
29°C
Partly cloudy
Search
ENG
IN
21:53
09-08-2026

ASSESSMENT TASK-2

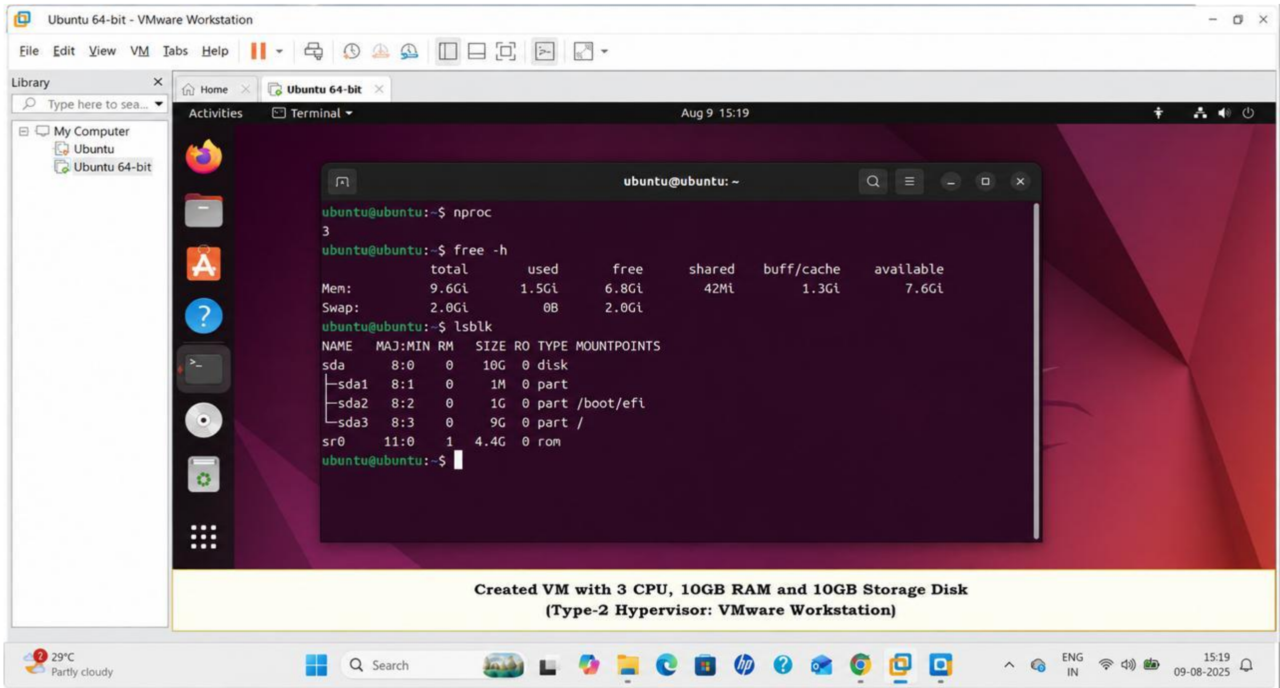
EXP 1:



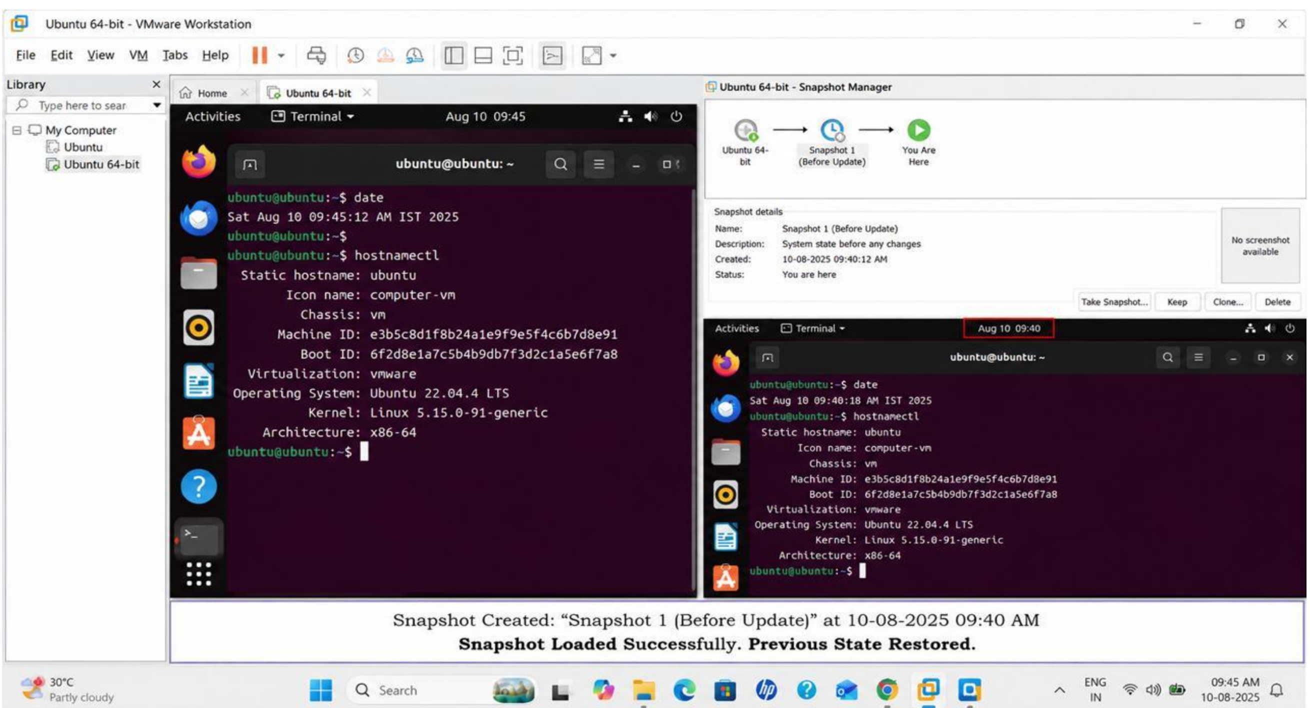
EXP 2:



EXP 3:

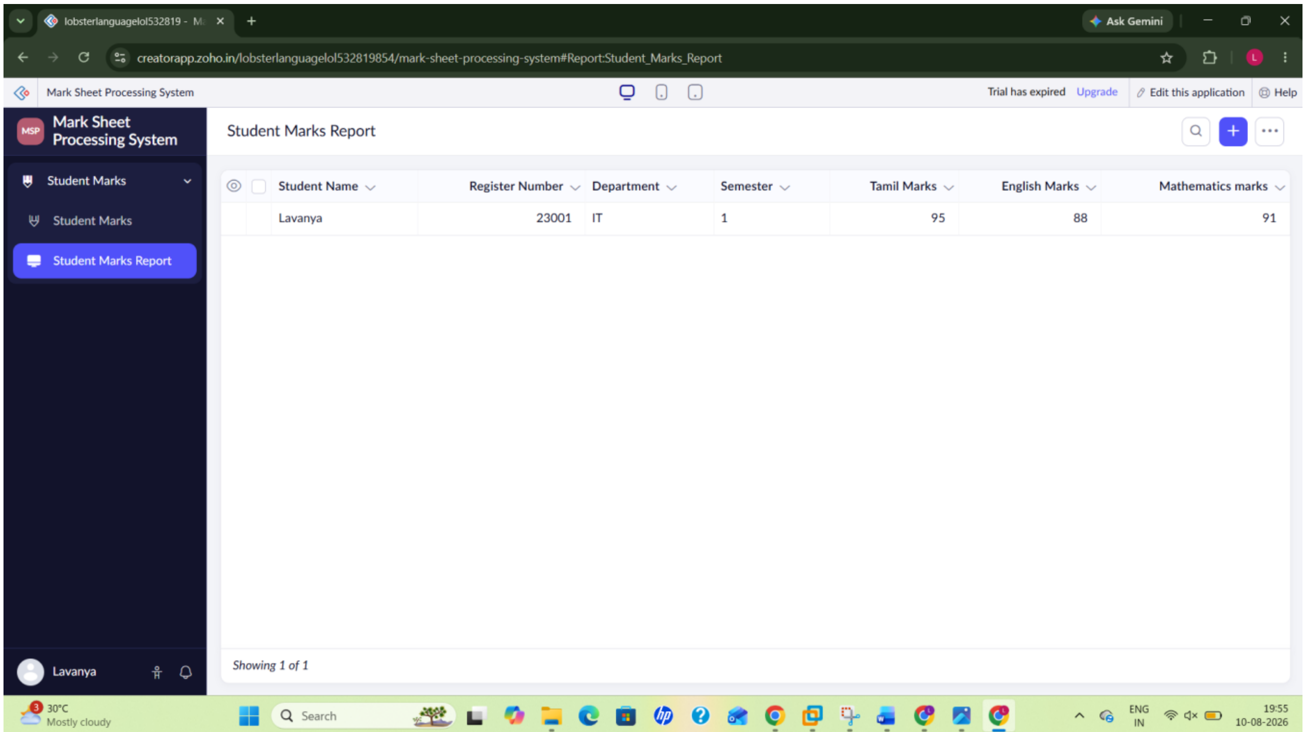

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu
Ubuntu 64-bit
Home
Ubuntu 64-bit
Activities
Terminal
Aug 9 15:19
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ nproc
3
ubuntu@ubuntu:~$ free -h
total used free shared buff/cache available
Mem: 9.6Gi 1.5Gi 6.8Gi 42Mi 1.3Gi 7.6Gi
Swap: 2.0Gi 0B 2.0Gi
ubuntu@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 10G 0 disk
├─sda1 8:1 0 1M 0 part
├─sda2 8:2 0 1G 0 part /boot/efi
└─sda3 8:3 0 9G 0 part /
sr0 11:0 1 4.4G 0 rom
ubuntu@ubuntu:~$
Created VM with 3 CPU, 10GB RAM and 10GB Storage Disk
(Type-2 Hypervisor: VMware Workstation)
29°C
Partly cloudy
Search
ENG
IN
15:19
09-08-2025

EXP 4:

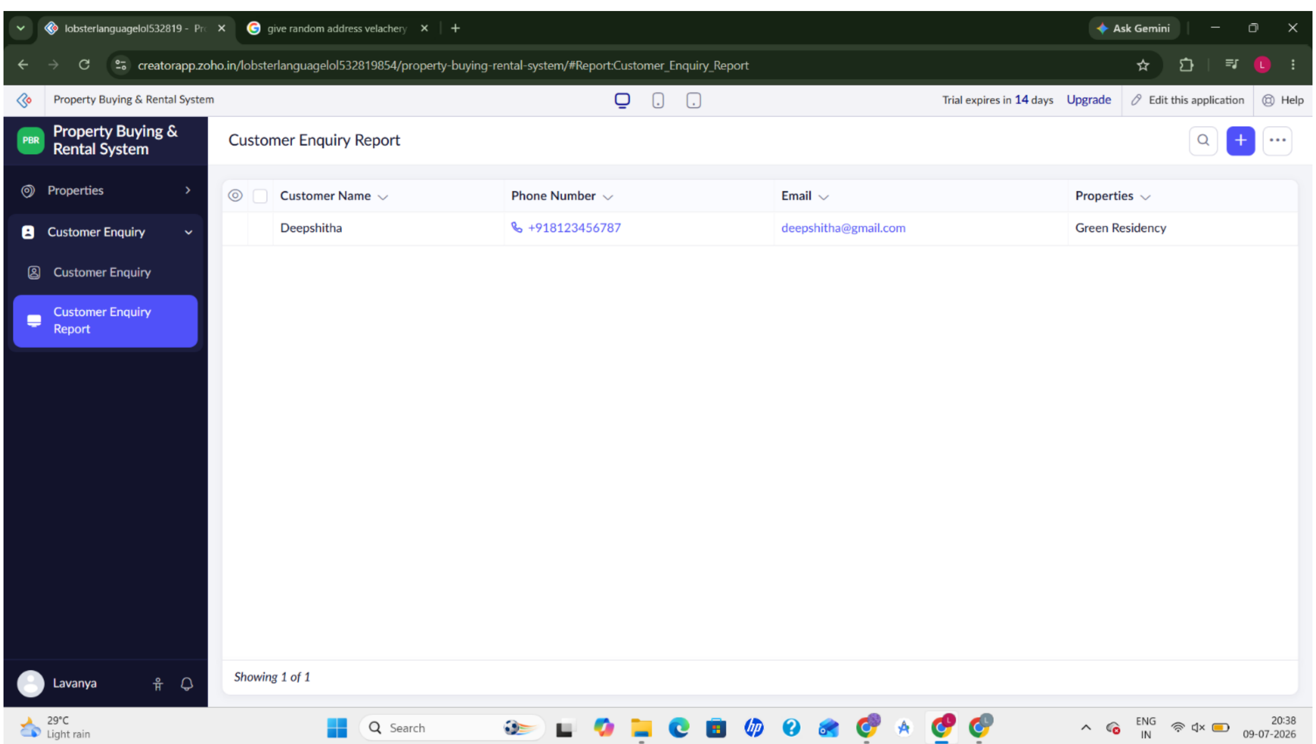

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu
Ubuntu 64-bit
Home
Ubuntu 64-bit
Activities
Terminal
Aug 10 09:45
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ date
Sat Aug 10 09:45:12 AM IST 2025
ubuntu@ubuntu:~$
ubuntu@ubuntu:~$ hostnamectl
Static hostname: ubuntu
Icon name: computer-vm
Chassis: vm
Machine ID: e3b5c8d1f8b24a1e9f9e5f4c6b7d8e91
Boot ID: 6f2d8e1a7c5b4b9db7f3d2c1a5e6f7a8
Virtualization: vmware
Operating System: Ubuntu 22.04.4 LTS
Kernel: Linux 5.15.0-91-generic
Architecture: x86-64
ubuntu@ubuntu:~$
Ubuntu 64-bit - Snapshot Manager
Ubuntu 64-bit
Snapshot 1 (Before Update)
You Are Here
Snapshot details
Name: Snapshot 1 (Before Update)
Description: System state before any changes
Created: 10-08-2025 09:40:12 AM
Status: You are here
No screenshot available
Take Snapshot... Keep Clone... Delete
Aug 10 09:40
ubuntu@ubuntu:~$ date
Sat Aug 10 09:40:18 AM IST 2025
ubuntu@ubuntu:~$ hostnamectl
Static hostname: ubuntu
Icon name: computer-vm
Chassis: vm
Machine ID: e3b5c8d1f8b24a1e9f9e5f4c6b7d8e91
Boot ID: 6f2d8e1a7c5b4b9db7f3d2c1a5e6f7a8
Virtualization: vmware
Operating System: Ubuntu 22.04.4 LTS
Kernel: Linux 5.15.0-91-generic
Architecture: x86-64
ubuntu@ubuntu:~$
Snapshot Created: "Snapshot 1 (Before Update)" at 10-08-2025 09:40 AM
Snapshot Loaded Successfully. Previous State Restored.
30°C
Partly cloudy
Search
ENG
IN
09:45 AM
10-08-2025

ASSESSMENT TASK 3

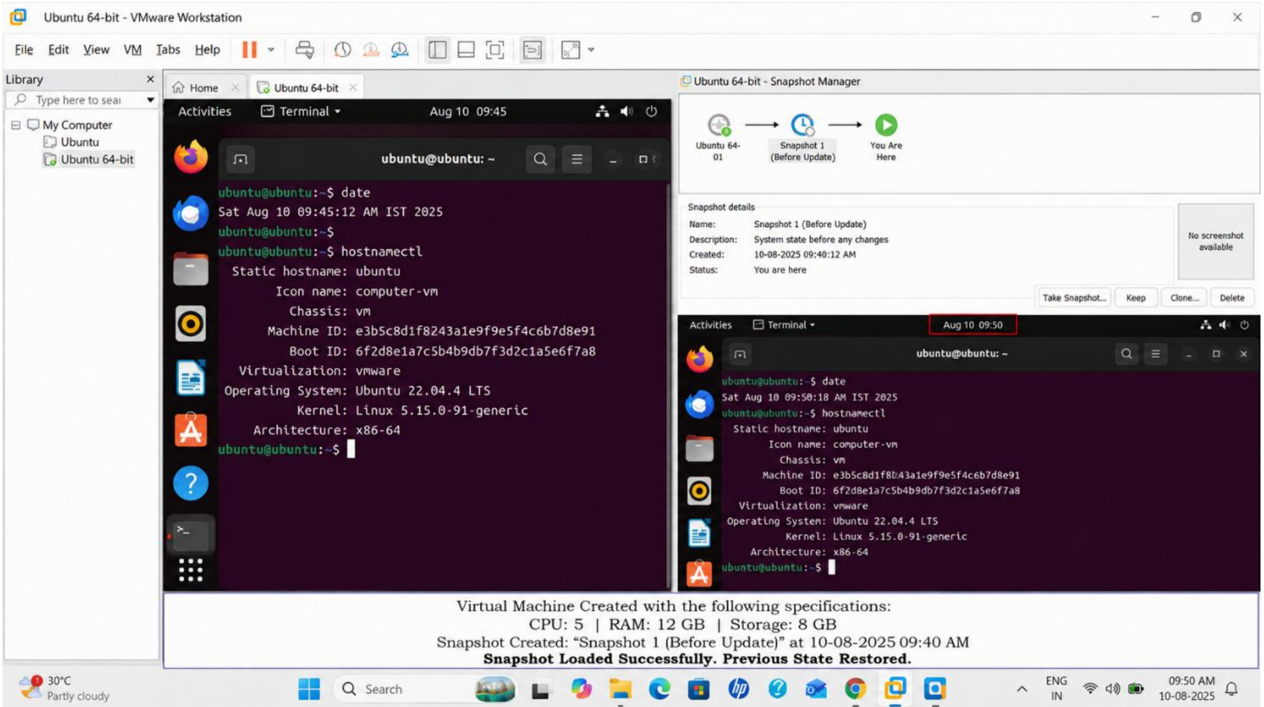
EXP 1:



EXP 2:

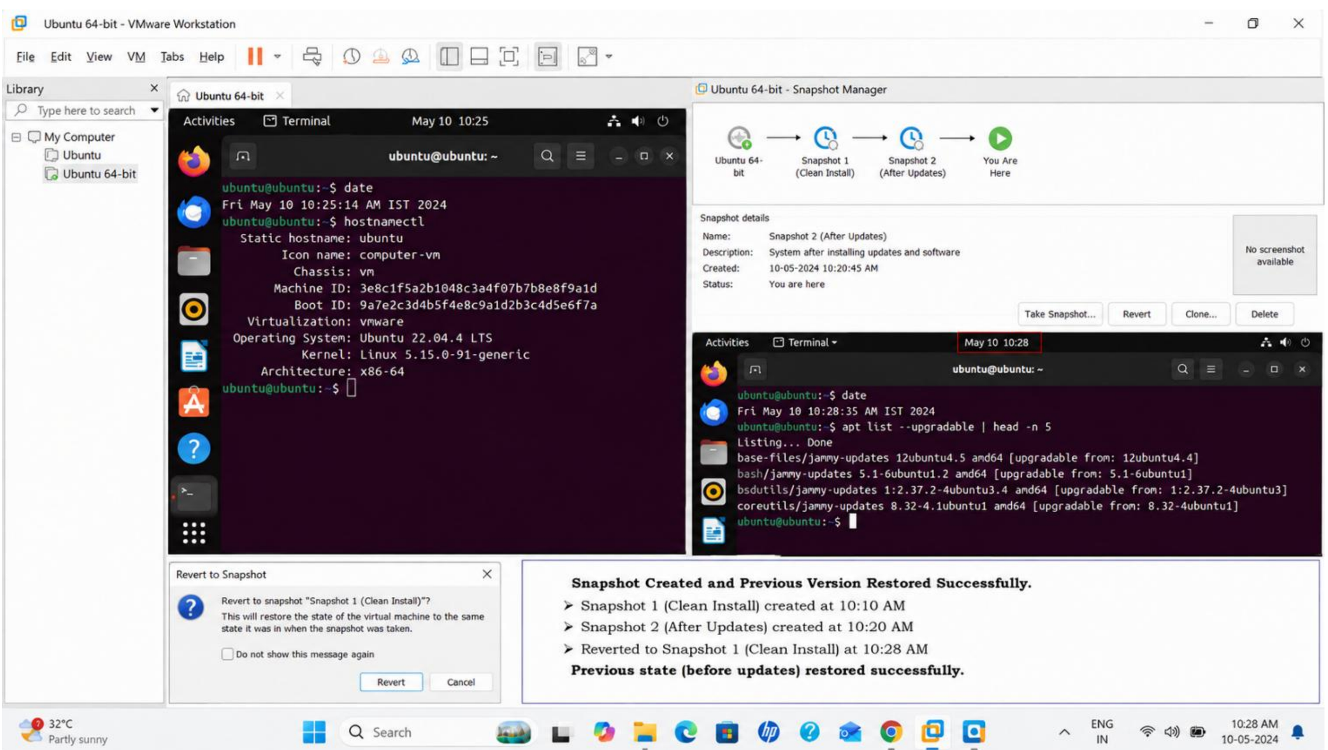


EXP 3:



Ubuntu 64-bit - VMware Workstation
Ubuntu 64-bit - Snapshot Manager
ubuntu@ubuntu:~$ date
Sat Aug 10 09:45:12 AM IST 2025
ubuntu@ubuntu:~$ hostnamectl
Static hostname: ubuntu
Icon name: computer-vm
Chassis: vm
Machine ID: e3b5c8d1f8243a1e9f9e5f4c6b7d8e91
Boot ID: 6f2d8e1a7c5b4b9db7f3d2c1a5e6f7a8
Virtualization: vmware
Operating System: Ubuntu 22.04.4 LTS
Kernel: Linux 5.15.0-91-generic
Architecture: x86-64
Snapshot 1 (Before Update)
You Are Here
Name: Snapshot 1 (Before Update)
Description: System state before any changes
Created: 10-08-2025 09:40:12 AM
Status: You are here
Virtual Machine Created with the following specifications:
CPU: 5 | RAM: 12 GB | Storage: 8 GB
Snapshot Created: "Snapshot 1 (Before Update)" at 10-08-2025 09:40 AM
Snapshot Loaded Successfully. Previous State Restored.

EXP 4:



Ubuntu 64-bit - VMware Workstation
Ubuntu 64-bit - Snapshot Manager
ubuntu@ubuntu:~$ date
Fri May 10 10:25:14 AM IST 2024
ubuntu@ubuntu:~$ hostnamectl
Static hostname: ubuntu
Icon name: computer-vm
Chassis: vm
Machine ID: 3e8c1f5a2b1048c3a4f07b7b8e8f9a1d
Boot ID: 9a7e2c3d4b5f4e8c9a1d2b3c4d5e6f7a
Virtualization: vmware
Operating System: Ubuntu 22.04.4 LTS
Kernel: Linux 5.15.0-91-generic
Architecture: x86-64
Snapshot 1 (Clean Install)
Snapshot 2 (After Updates)
You Are Here
Name: Snapshot 2 (After Updates)
Description: System after installing updates and software
Created: 10-05-2024 10:20:45 AM
Status: You are here
ubuntu@ubuntu:~$ apt list --upgradable | head -n 5
Revert to Snapshot
Revert to snapshot "Snapshot 1 (Clean Install)"?
This will restore the state of the virtual machine to the same state it was in when the snapshot was taken.
Snapshot Created and Previous Version Restored Successfully.
Snapshot 1 (Clean Install) created at 10:10 AM
Snapshot 2 (After Updates) created at 10:20 AM
Reverted to Snapshot 1 (Clean Install) at 10:28 AM
Previous state (before updates) restored successfully.